

Chaithanya Virupaksha

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Eligible to work in the UK on the Graduate Route (2 years) — available immediately

PROFILE

Software engineer and MSc Advanced Computer Science candidate at Newcastle University, with two years' industry experience before returning to study. Builds backend and cloud-native systems in Java and Python, and ships production front ends in React. Recent work includes a dissertation characterising federated learning under wireless packet loss, OAuth2/Keycloak authentication against a Kubernetes-hosted microservices backend on AKS, and front-end delivery in a commercial agile team. Comfortable in client-facing settings and translating technical constraints into decisions non-engineers can act on. Seeking a graduate software engineering role with a cloud or platform focus, available to start immediately.

TECHNICAL SKILLS

Languages: Java 21, Python, C, C++, JavaScript (ES6+), SQL

Backend: Spring Boot 3, Flask, REST/OpenAPI, microservices

Cloud & DevOps: Docker, Kubernetes/AKS, GitHub Actions (CI/CD), NGINX, Git; AWS Solutions Architect Associate in progress

Front end: React 19, Vite, React Router, HTML5, SCSS, Bootstrap

Data & ML: PostgreSQL, MySQL, Pandas, scikit-learn, PyTorch, NumPy; distributed training, experimental design

Security: OAuth2/OIDC, JWT handling, Keycloak, threat modelling, NIST risk frameworks

PROJECTS

MSc Dissertation — Federated Learning under Wireless Packet Loss (Python, PyTorch) | 2026

CSC8499 Individual Research Project · Supervisor: Dr Rehmat Ullah, Newcastle University

- Built a distributed-training simulator instrumented per client per round, separating communication failure from learning failure — the design decision that made every downstream result attributable to a mechanism rather than to average accuracy.
- Modelled three distinct network failure modes (independent loss, Gilbert-Elliott burst loss, bandwidth-constrained deadline failure) across a factorial of compression schemes, four client-selection policies and IID/non-IID data partitions.
- Used common random numbers via a keyed cryptographic digest so arms differing only in algorithm saw identical channel realisations, removing channel noise from every paired comparison.
- Found and reported the central result that gradient compression changes which clients can participate at all: the accuracy spread across selection policies narrowed from 16.4 to 2.3 percentage points under Top-k sparsification.
- Identified a defect in the contribution estimator during a post-hoc audit, re-ran the affected arms under corrected observability, and reported the measured difference rather than silently patching it.
- 50 unit tests across independently testable channel, selection and mitigation modules; all tables and figures generated from logged CSV/JSON output, with no hand-transcribed numbers.

Cloud-Native Bookstore Platform — React 19, Keycloak, Spring Boot on AKS | 2025–26

MSc group project, team of 6 · github.com/paritosh4495/CSC8113_CloudComputing_Bookstore

- Built front-end features for a React 19 / Vite SPA consuming a Spring Boot microservices backend, working across the catalogue, book detail, cart and checkout pages.
- Integrated browser-side authentication with Keycloak using keycloak-js: parsed JWT claims into application state, handled silent token refresh on expiry, and gated authenticated routes and admin-only actions on realm roles.
- Implemented optimistic cart updates in React context — applying quantity and removal changes locally before the API responded, with re-fetch on failure to recover consistency — trading a brief risk of divergence for a noticeably more responsive UI.
- Worked within a team whose wider platform covered AKS deployment, blue-green releases, HPA auto-scaling and Velero disaster recovery; co-authored the group technical report documenting the architecture.

AI Bias Audit Framework — Python, scikit-learn, NLP | 2026

Individual project

- Built a Python framework quantifying bias in LLM-generated text across gender, cultural and occupational-stereotype dimensions, comparing output from multiple models.
- Applied TF-IDF, LDA topic modelling and Shannon entropy to produce reproducible diversity metrics, with visualisations supporting comparison across models.

Tokeneer Security Risk Assessment — NIST, threat modelling | 2026

Individual coursework · report available on request

- Assessed the Tokeneer secure-enclave system against NIST risk frameworks, modelling threats including biometric spoofing and privilege escalation, and prioritising mitigations by business impact.

EXPERIENCE

Technology Consulting Analyst — Accurate InfoTech Solutions, Bengaluru | May 2024 – Aug 2025

- Advised academic and enterprise clients on system architecture and process improvement, running technical assessments and gap analyses and presenting recommendations to non-technical stakeholders.
- Coordinated partnership programmes with engineering institutions, managing delivery timelines and acting as the technical point of contact across multiple concurrent engagements.
- Learned to surface underlying requirements early — several engagements changed scope materially once the real need was questioned rather than assumed.

Full-Stack Development Intern — PACE Wisdom, Bengaluru | Sep 2023 – Mar 2024

- Built responsive, component-based UI in HTML, SCSS and Bootstrap, improving the Lighthouse performance score of the pages worked on by roughly 25%.
- Worked in a two-week agile cycle alongside design and back-end engineers, taking features from ticket through code review to release.

Machine Learning Intern — Accurate InfoTech Solutions, Bengaluru | Dec 2022 – Jan 2023

- Built an object-detection pipeline in Python with OpenCV covering several object categories, handling image pre-processing and evaluation.
- Prototyped a CNN classifier for plant-disease identification from leaf images, benchmarked against a rule-based baseline.

EDUCATION

MSc Advanced Computer Science — Newcastle University | 2025 – Dec 2026

Modules: Cloud Computing · Secure Software Development · Risk & Trust Management · Enterprise Software & Services · Advanced Programming in Java · Generative AI for Businesses

BEng Computer Science — Dr. Ambedkar Institute of Technology, Bengaluru | Jul 2024

CGPA 8.3/10 (equivalent to a UK First / high 2:1)

PUBLICATIONS & ADDITIONAL

- C. Virupaksha, “Privacy-Preserving Digital Twin Models for Smart Healthcare Applications,” IJDTSC, 2(1), 2026.
- Dr. Siddaraju, C. Virupaksha et al., “Smart Irrigation Using Embedded AI,” IJIRCCE, 12(6), 2024.
- Organised Kalarava 2024, a college-level festival, coordinating volunteers and logistics across multiple events.
- Happy to travel within the UK for client engagements.